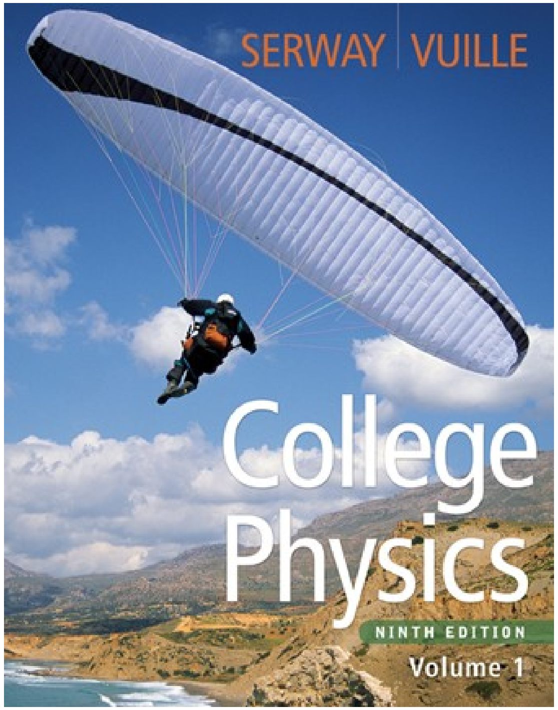

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Chris Vuille

Chapter Twelve

The Laws of Thermodynamics

First Law of Thermodynamics

- The First Law of Thermodynamics tells us that the internal energy of a system can be increased by
  - Adding energy to the system
  - Doing work on the system
- There are many processes through which these could be accomplished
  - As long as energy is conserved

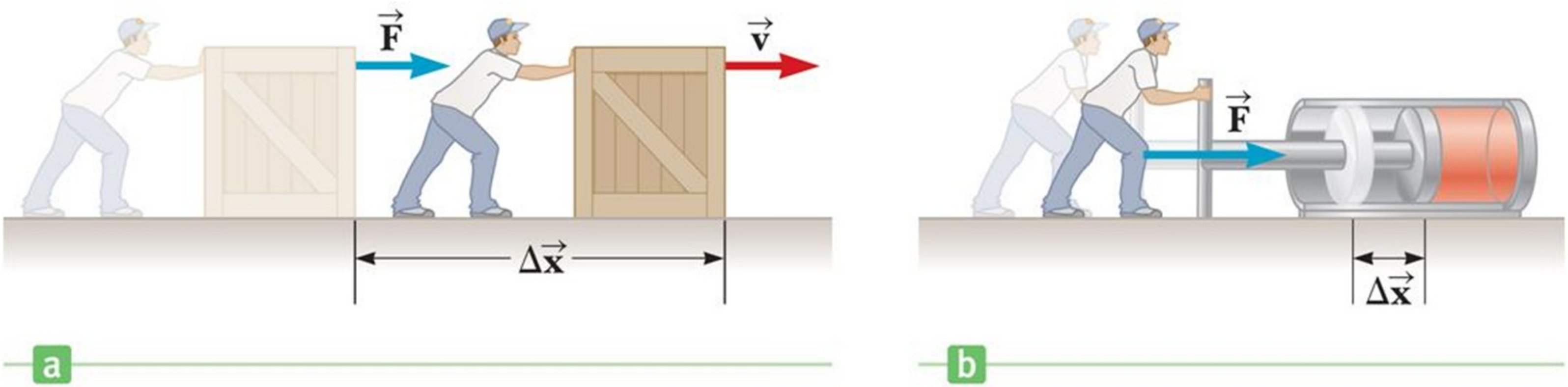
Second Law of Thermodynamics

- Constrains the First Law
- Establishes which processes actually occur
- Heat engines are an important application

Work in Thermodynamic Processes – Assumptions

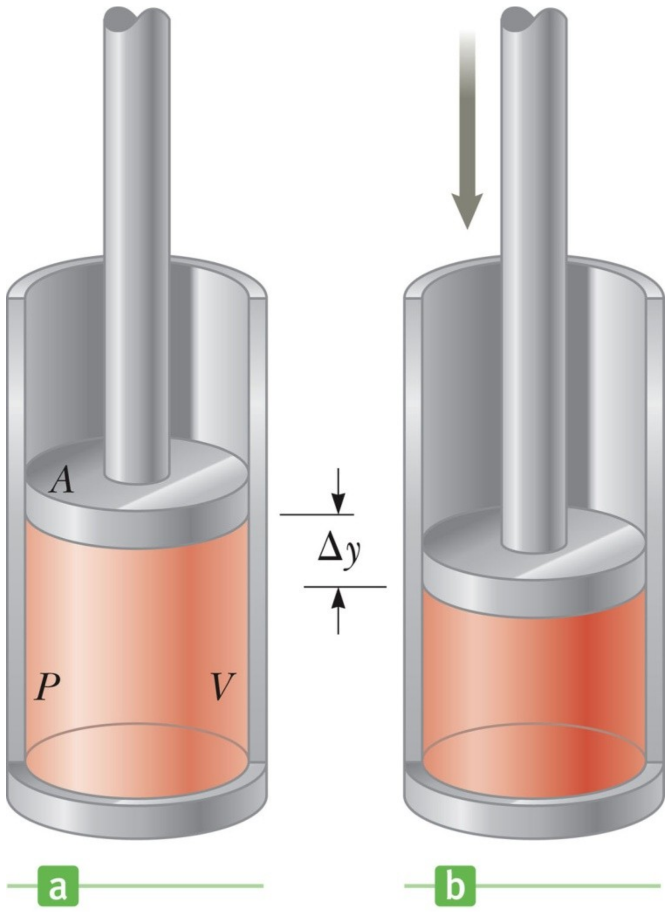
- Dealing with a gas
- Assumed to be in thermodynamic equilibrium
  - Every part of the gas is at the same temperature
  - Every part of the gas is at the same pressure
- Ideal gas law applies

Work in Thermodynamic Processes (2 of 5)



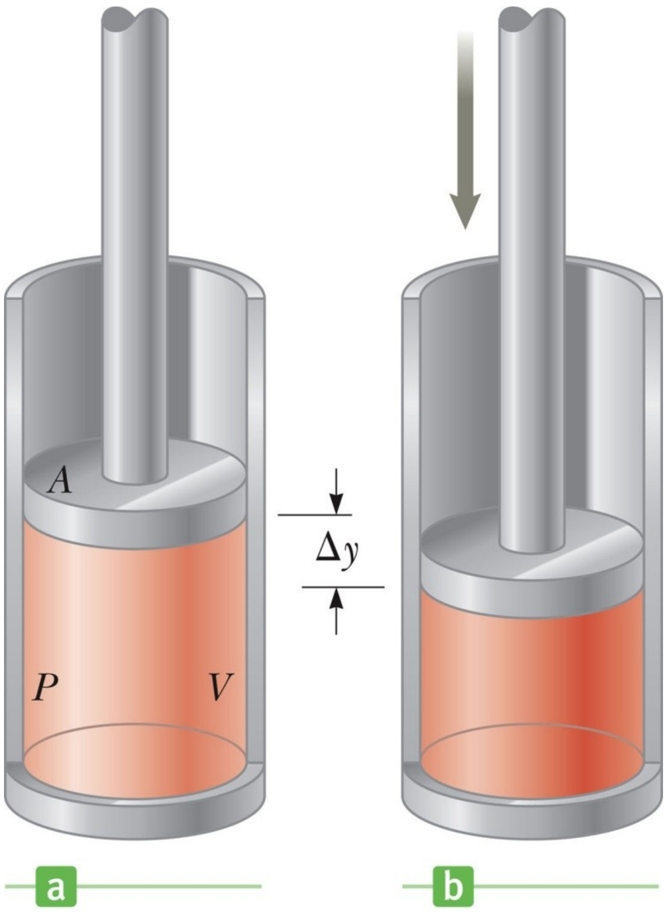
Work in a Gas Cylinder

- The gas is contained in a cylinder with a moveable piston
- The gas occupies a volume V and exerts pressure P on the walls of the cylinder and on the piston



Work in a Gas Cylinder, cont.

- A force is applied to slowly compress the gas
  - The compression is slow enough for all the system to remain essentially in thermal equilibrium
- $W = - P \Delta V$
  - This is the work done *on* the gas where P is the pressure throughout the gas



More about Work on a Gas Cylinder

- When the gas is compressed
  - ΔV is negative
  - The work done on the gas is positive
- When the gas is allowed to expand
  - ΔV is positive
  - The work done on the gas is negative
- When the volume remains constant
  - No work is done on the gas

Work By vs. Work On

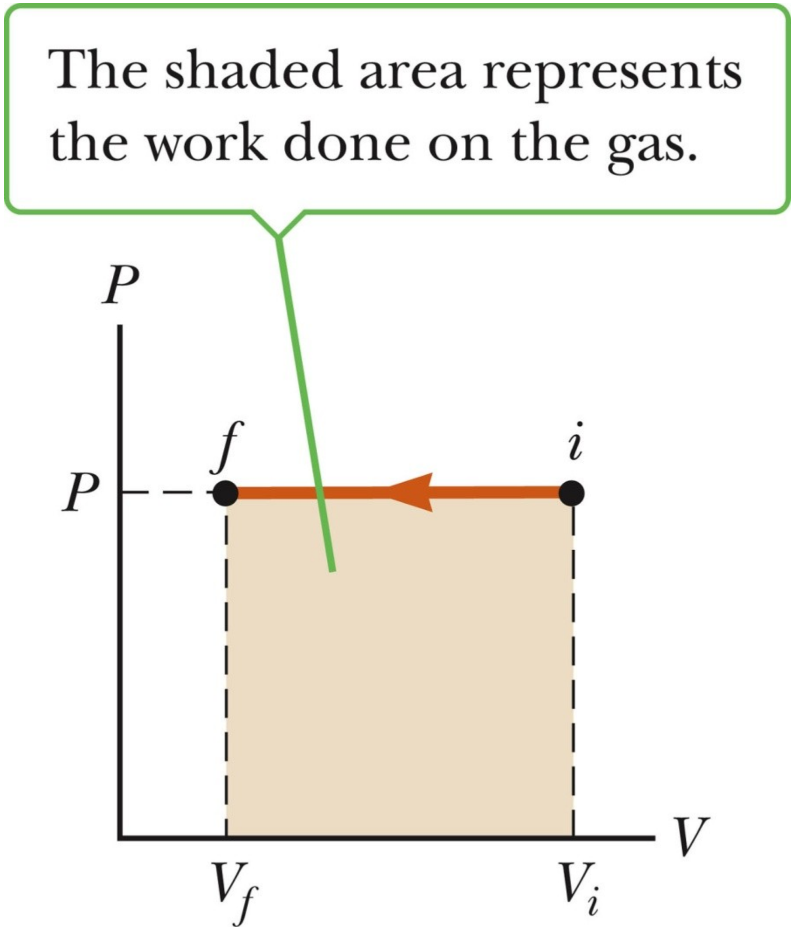
- The definition of work, W, specifies the work done **on** the gas (as opposed to: by the gas)
  - This definition focuses on the internal energy of the system
- W_{env} is used to denote the work done **by** the gas (on the environment)
  - The focus would be on harnessing a system's internal energy to do work on something external to the gas
- $W = - W_{env}$

Notes about the Work Equation

- The pressure remains constant during the expansion or compression
  - This is called an *isobaric* process
- The previous work equation can be used only for an isobaric process

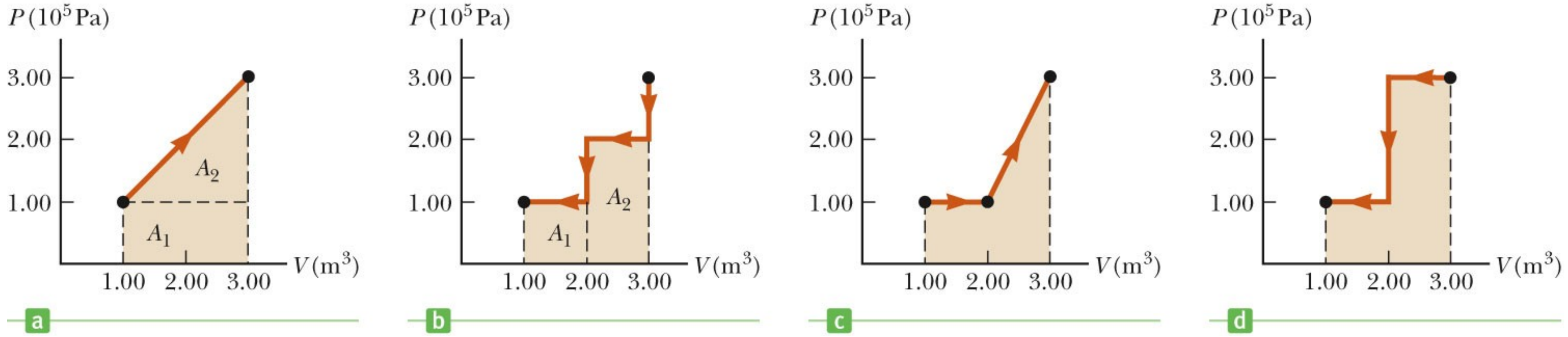
PV Diagrams

- Used when the pressure and volume are known at each step of the process
- The work done on a gas that takes it from some initial state to some final state is equal in magnitude to the area under the curve on the PV diagram
  - This is true whether or not the pressure stays constant



PV Diagrams, cont.

- The curve on the diagram is called the *path* taken between the initial and final states
- The work done depends on the particular path
  - Same initial and final states, but different amounts of work are done



First Law of Thermodynamics

- Energy conservation law
- Relates changes in internal energy to energy transfers due to heat and work
- Applicable to all types of processes
- Provides a connection between microscopic and macroscopic worlds

First Law, cont.

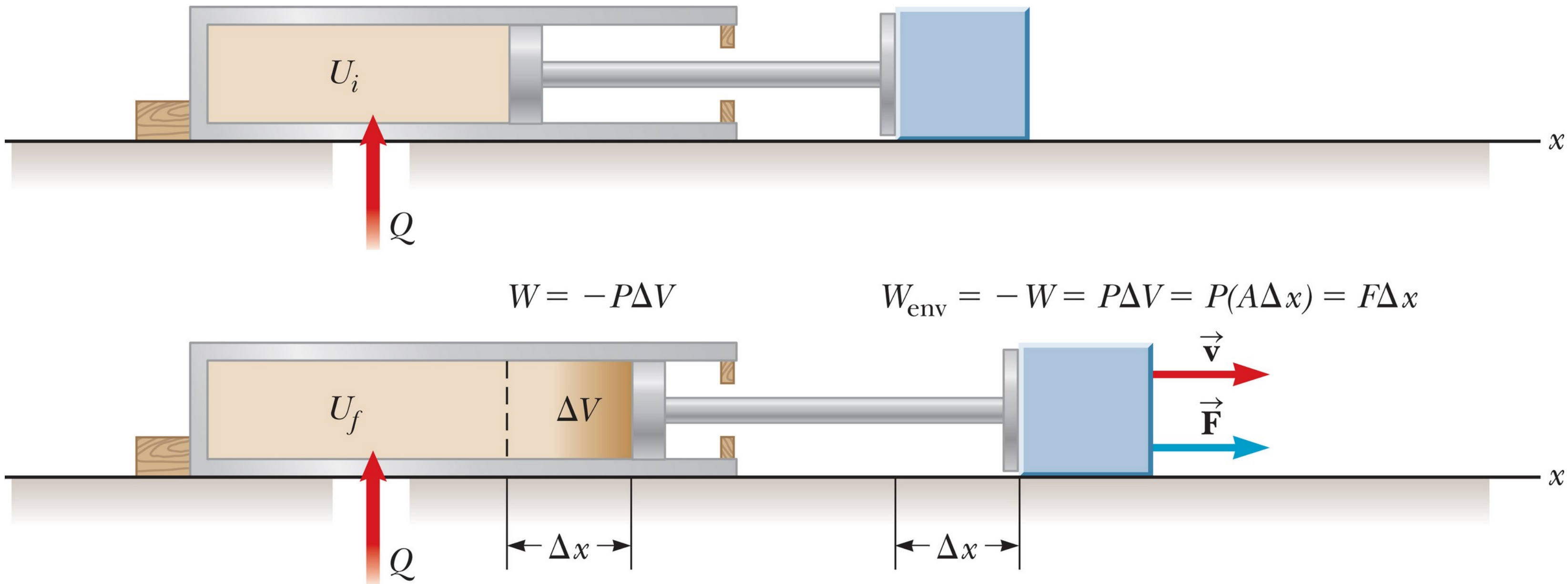
- Energy transfers occur
  - By doing work
    - Requires a macroscopic displacement of an object through the application of a force
  - By heat
    - Due to a temperature difference
    - Usually occurs by radiation, conduction and/or convection
  - Other methods are possible
- All result in a change in the internal energy, ΔU, of the system

The First Law of Thermodynamics (2 of 3)

First Law of Thermodynamics:

$$\Delta U = U_f - U_i = Q + W$$

The First Law of Thermodynamics (3 of 3)



$$\Delta U = U_f - U_i = Q + W$$

First Law, Equation

- If a system undergoes a change from an initial state to a final state, then $\Delta U = U_f - U_i = Q + W$
  - Q is the energy transferred between the system and the environment
  - W is the work done on the system
  - ΔU is the change in internal energy

First Law – Signs

- Signs of the terms in the equation
  - Q
    - Positive if energy is transferred *into* the system
    - Negative if energy is *removed from* the system
  - W
    - Positive if work is done *on* the system
    - Negative if work is done *by* the system
  - ΔU
    - Positive if the temperature increases
    - Negative if the temperature decreases

Notes About Work

- Positive work increases the internal energy of the system
- Negative work decreases the internal energy of the system
- This is consistent with the definition of mechanical work

Molar Specific Heat

- The molar specific heat at constant volume for a monatomic ideal gas
  - $C_v = 3/2\ R$
- The change in internal energy can be expressed as $\Delta U = n\ C_v\ \Delta T$
  - For an ideal gas, this expression is always valid, even if not at a constant volume

Molar Specific Heat, cont

- A gas with a larger molar specific heat requires more energy for a given temperature change
- The value depends on the structure of the gas molecule
- The value also depends on the ways the molecule can store energy

Work in Thermodynamic Processes (5 of 5)

Table 12.1 Molar Specific Heats of Various Gases

Gas	Molar Specific Heat (J/mol · K)[a]			
	C_p	C_v	$C_p - C_v$	$\gamma = C_p/C_v$
Monatomic Gases				
He	20.8	12.5	8.33	1.67
Ar	20.8	12.5	8.33	1.67
Ne	20.8	12.7	8.12	1.64
Kr	20.8	12.3	8.49	1.69
Diatomic Gases				
H_2	28.8	20.4	8.33	1.41
N_2	29.1	20.8	8.33	1.40
O_2	29.4	21.1	8.33	1.40
CO	29.3	21.0	8.33	1.40
Cl_2	34.7	25.7	8.96	1.35
Polyatomic Gases				
CO_2	37.0	28.5	8.50	1.30
SO_2	40.4	31.4	9.00	1.29
H_2O	35.4	27.0	8.37	1.30
CH_4	35.5	27.1	8.41	1.31

[a]All values except that for water were obtained at 300 K.

Degrees of Freedom

- Each way a gas can store energy is called a *degree of freedom*
- Each degree of freedom contributes ½ R to the molar specific heat
- See table 12.1 for some C_v values

Types of Thermal Processes

- *Isobaric*
  - Pressure stays constant, $\Delta P = 0$
  - Horizontal line on the PV diagram
- *Adiabatic*
  - No heat is exchanged with the surroundings, $Q = 0$
- *Isovolumetric*
  - Volume stays constant, $W = 0$
  - Vertical line on the PV diagram
- *Isothermal*
  - Temperature stays the same $\Delta T = 0$, $\Delta U = 0$

Isobaric Processes

- The work done by an expanding gas in an isobaric process is at the expense of the internal energy of the gas
- $Q = 5/2\ n\ R\ \Delta T = n\ C_P\ \Delta T$
  - C_p is the molar heat capacity at constant pressure
  - $C_P = C_v + R$

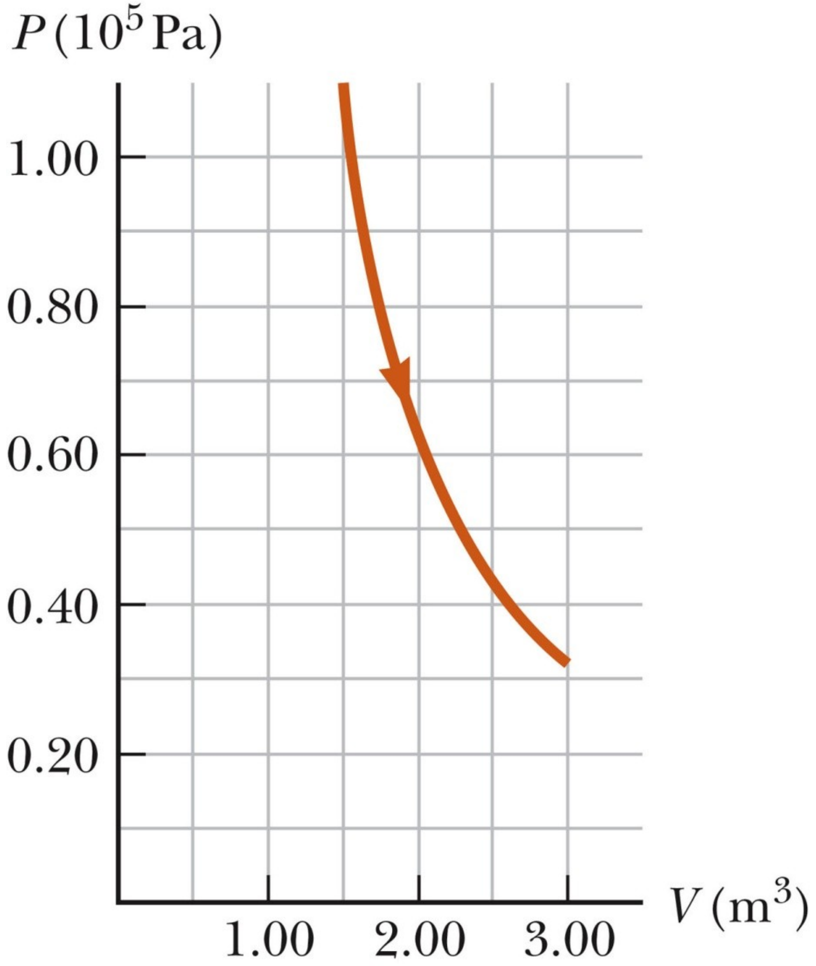
Adiabatic Processes

- For an adiabatic process, Q = 0
- First Law becomes $\Delta U = W$
- For an ideal gas undergoing an adiabatic process

$$PV^{\gamma} = constant \quad where \quad \gamma = \frac{C_P}{C_V}$$

  - γ is called the *adiabatic index* of the gas

Adiabatic Expansion, Diagram



Section 12.3

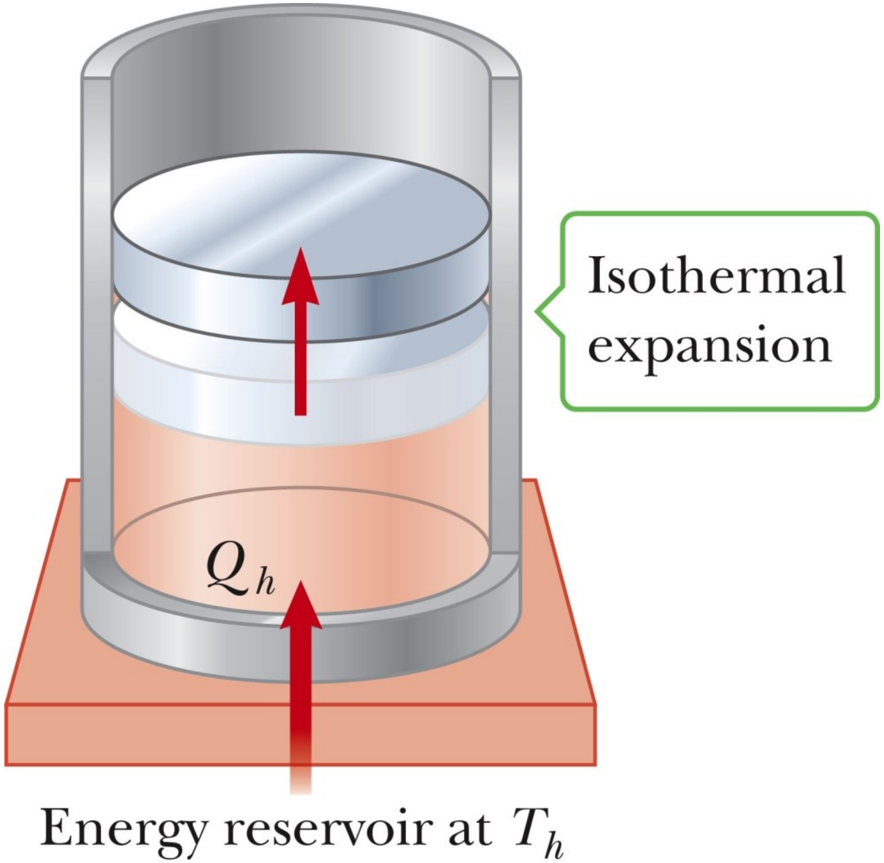
Isovolumetric Process

- Also called *isochoric* process
- Constant volume
  - Vertical line on PV diagram
- $W = 0$ (since $\Delta V = 0$)
- First Law becomes $\Delta U = Q$
  - The change in internal energy equals the energy transferred to the system by heat
- $Q = n C_v \Delta T$

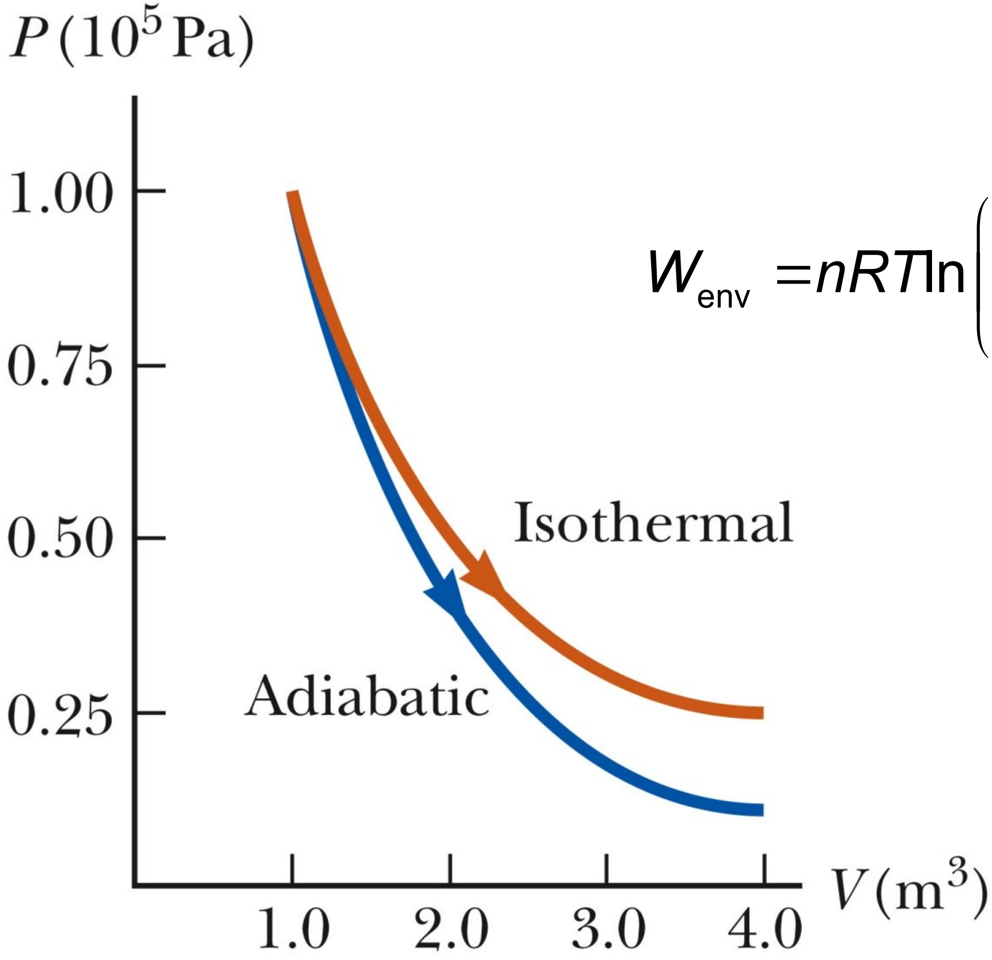
Isothermal Process

- The temperature doesn't change
  - In an ideal gas, since $\Delta T = 0$, the $\Delta U = 0$
- First Law becomes $W = - Q$ and

$$W_{env} = nRT\, ln\left(\frac{V_f}{V_i}\right)$$



Isothermal Process, Diagram



$$W_{\text{env}} = nRT \ln\left(\frac{V_f}{V_i}\right)$$

General Case

- Can still use the First Law to get information about the processes
- Work can be computed from the PV diagram
- If the temperatures at the endpoints can be found, ΔU can be found

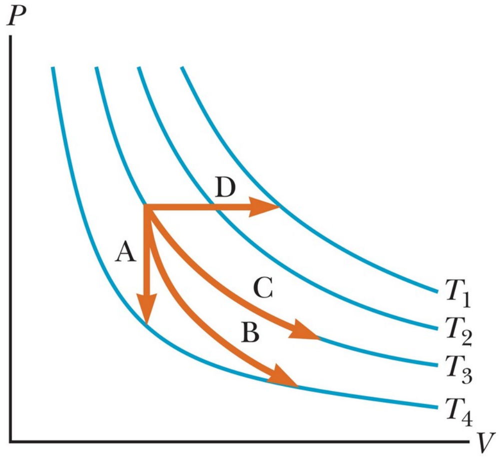
Summary of Processes

Table 12.2 The First Law and Thermodynamic Processes (Ideal Gases)

Process	ΔU	Q	W
Isobaric	$nC_v\,\Delta T$	$nC_p\,\Delta T$	$-P\,\Delta V$
Adiabatic	$nC_v\,\Delta T$	0	ΔU
Isovolumetric	$nC_v\,\Delta T$	ΔU	0
Isothermal	0	$-W$	$-nRT\ln\left(\frac{V_f}{V_i}\right)$
General	$nC_v\,\Delta T$	$\Delta U - W$	(PV Area)

Think – Pair – Share 2

What are the different paths in the figure?



1. A: isobaric; B: isothermal; C: isovolumetric; D: adiabatic
2. A: isovolumetric; B: adiabatic; C: isothermal; D: isobaric
3. A: adiabatic; B: isobaric; C: isothermal; D: isovolumetric

Heat Engine

- A heat engine takes in energy by heat and partially converts it to other forms
- In general, a heat engine carries some working substance through a cyclic process

Cyclic Process in a Heat Engine

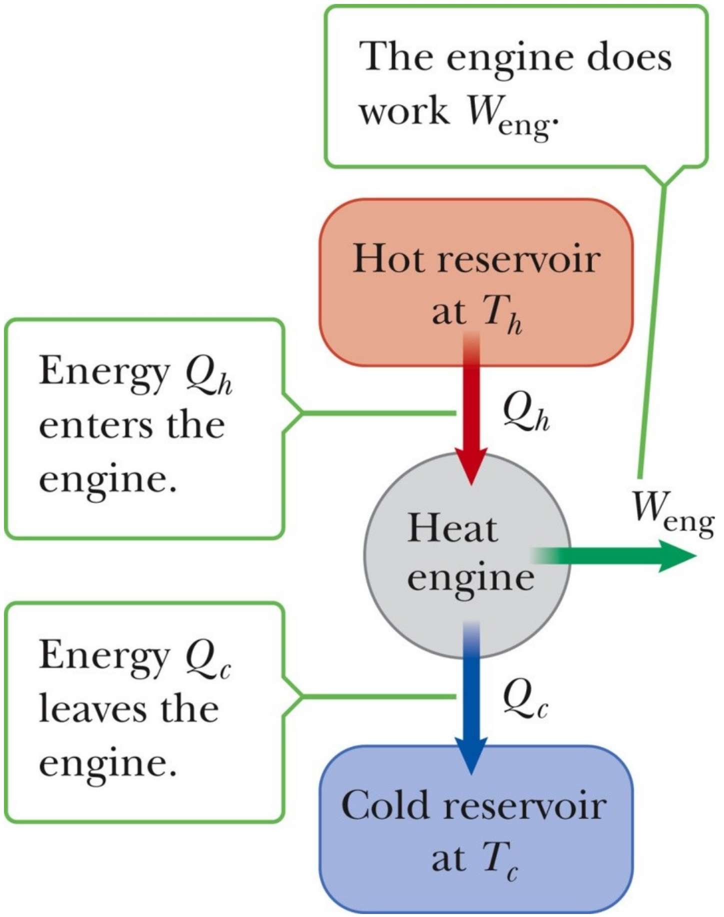
- Energy is transferred by heat from a source at a high temperature
- Work is done by the engine
- Energy is expelled by the engine by heat to a source at a lower temperature

Cyclic Processes

- The process originates and ends at the same state
  - $U_f = U_i$ and $Q = -W = W_{eng}$
- The net work done per cycle by the gas is equal to the area enclosed by the path representing the process on a PV diagram

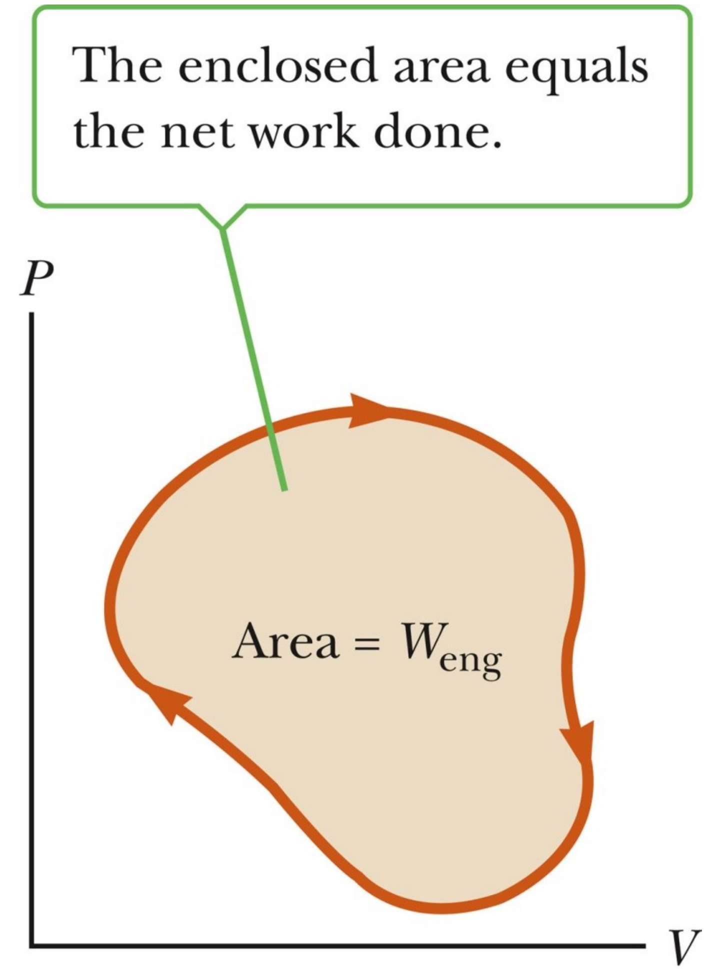
Heat Engine, cont.

- Energy is transferred from a source at a high temperature (Q_h)
- Work is done by the engine (W_{eng})
- Energy is expelled to a source at a lower temperature (Q_c)



Heat Engine, cont.

- Since it is a cyclical process, $\Delta U = 0$
  - Its initial and final internal energies are the same
- Therefore, $Q_{net} = W_{eng}$
- The work done by the engine equals the net energy absorbed by the engine
- The work is equal to the area enclosed by the curve of the PV diagram



Thermal Efficiency of a Heat Engine

- Thermal efficiency is defined as the ratio of the work done by the engine to the energy absorbed at the higher temperature

$$e \equiv \frac{W_{eng}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|}$$

- e = 1 (100% efficiency) only if $Q_c = 0$
  - No energy expelled to cold reservoir

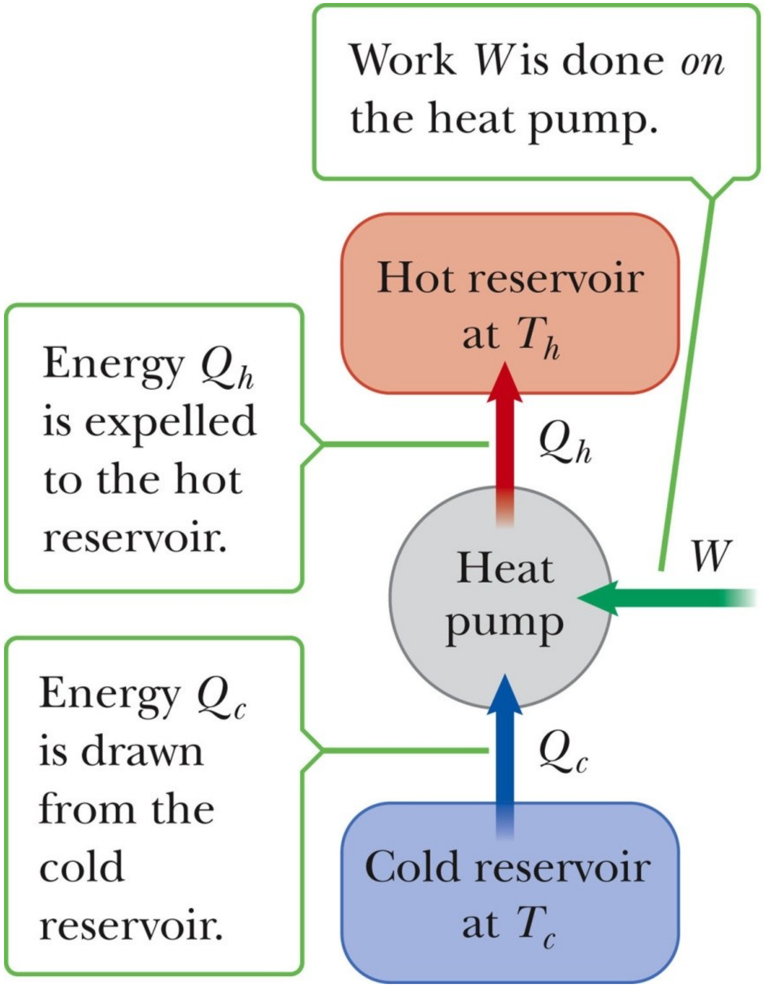
Example:

Refrigerators and Heat Pumps

- Heat engines can operate in reverse
  - Energy is injected
  - Energy is extracted from the cold reservoir
  - Energy is transferred to the hot reservoir
- This process means the heat engine is running as a heat pump
  - A refrigerator is a common type of heat pump
  - An air conditioner is another example of a heat pump

Heat Pump, cont

- The work is what you pay for
- The Q_c is the desired benefit
- The coefficient of performance (COP) measures the performance of the heat pump running in cooling mode



Heat Pumps and Refrigerators

Heat Pump, COP

- In cooling mode,

$$COP = \frac{|Q_c|}{W}$$

- The higher the number, the better
- A good refrigerator or air conditioner typically has a COP of 5 or 6

Heat Pump, COP

- In heating mode,

$$COP = \frac{|Q_H|}{W}$$

- The heat pump warms the inside of the house by extracting heat from the colder outside air
- Typical values are greater than one

William Thomson, Lord Kelvin

- 1824 – 1907
- British physicist
- First to propose the use of an absolute temperature scale
- Formulated a version of the Second Law

Second Law of Thermodynamics

- No heat engine operating in a cycle can absorb energy from a reservoir and use it entirely for the performance of an equal amount of work
  - Kelvin – Planck statement
  - Means that Q_c cannot equal 0
    - Some Q_c must be expelled to the environment
  - Means that e must be less than 100%

Summary of the First and Second Laws

- First Law
  - We cannot get a greater amount of energy out of a cyclic process than we put in
- Second Law
  - We can't break even

Second Law, Alternative Statement

- If two systems are in thermal contact, net thermal energy transfers spontaneously by heat from the hotter system to the colder system
  - The heat transfer occurs without work being done

Reversible and Irreversible Processes

- A *reversible* process is one in which every state along some path is an equilibrium state
  - The system can return to its initial state along the same path
- An *irreversible* process does not meet these requirements
  - Most natural processes are irreversible
- Reversible processes are an idealization, but some real processes are good approximations

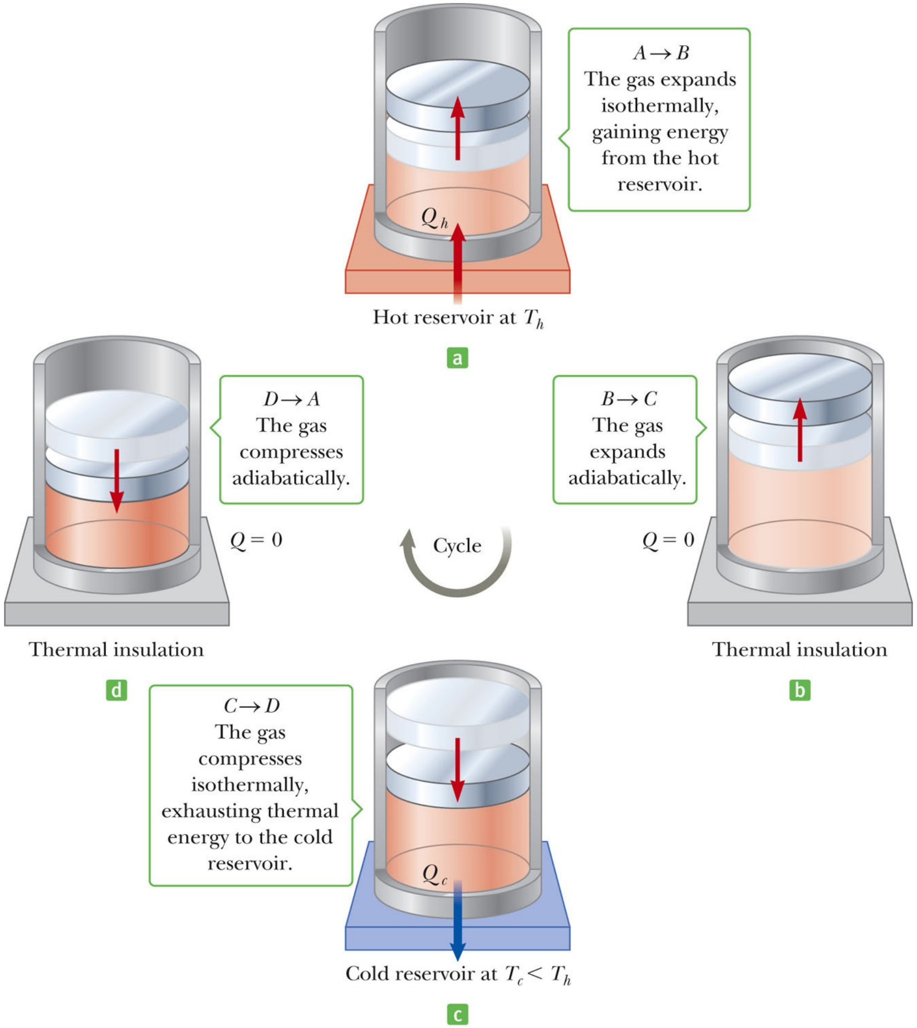
Sadi Carnot

- 1796 – 1832
- French Engineer
- Founder of the science of thermodynamics
- First to recognize the relationship between work and heat

Carnot Engine

- A theoretical engine developed by Sadi Carnot
- A heat engine operating in an ideal, reversible cycle (now called a *Carnot Cycle*) between two reservoirs is the most efficient engine possible
- *Carnot's Theorem*: No real engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs

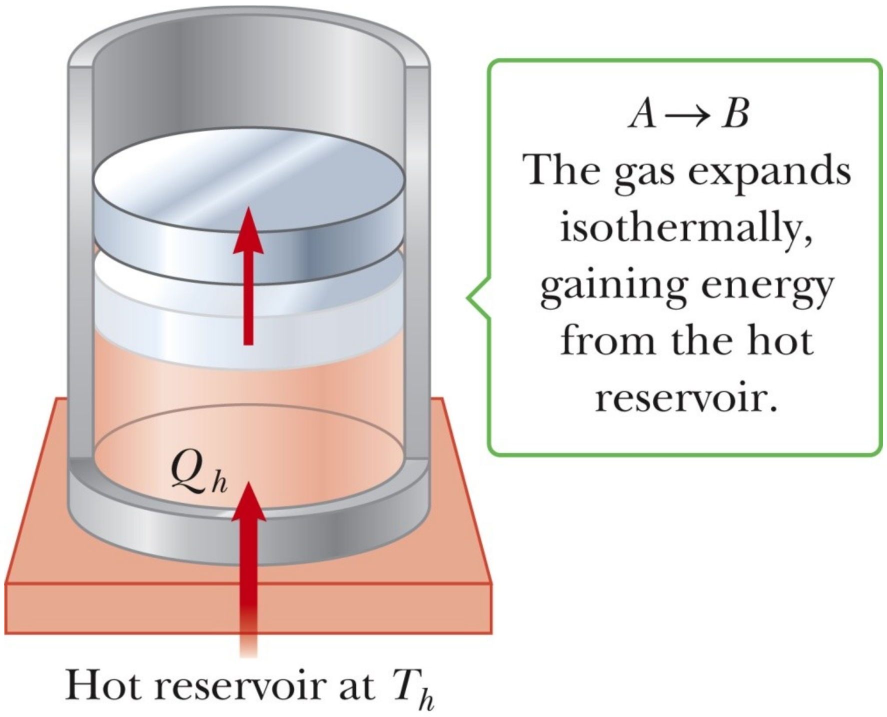
Carnot Cycle



Section 12.4

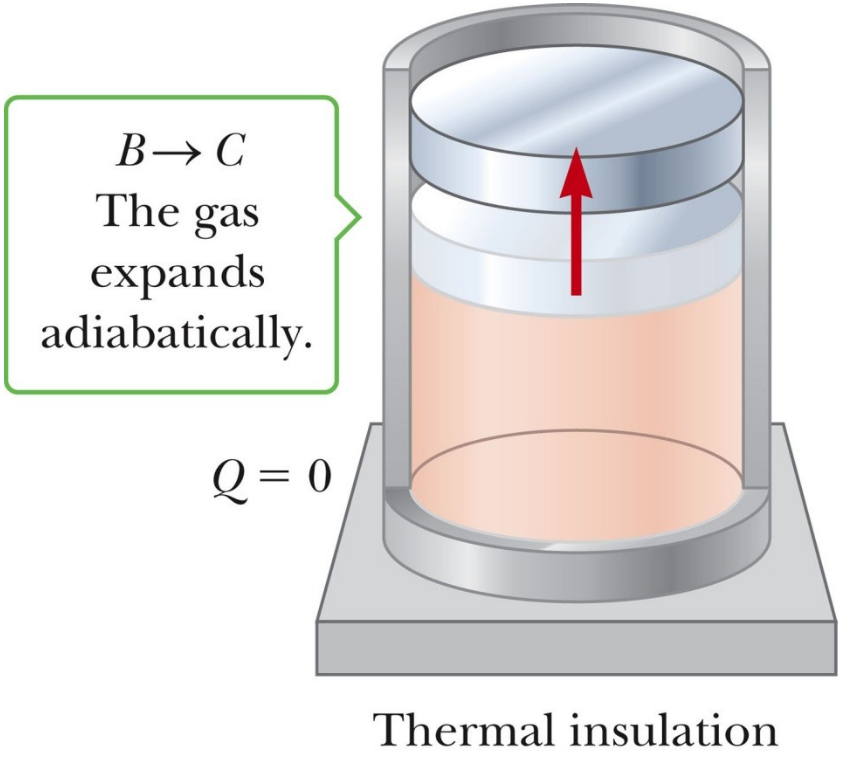
Carnot Cycle, A to B

- A to B is an isothermal expansion at temperature T_h
- The gas is placed in contact with the high temperature reservoir
- The gas absorbs heat Q_h
- The gas does work W_{AB} in raising the piston



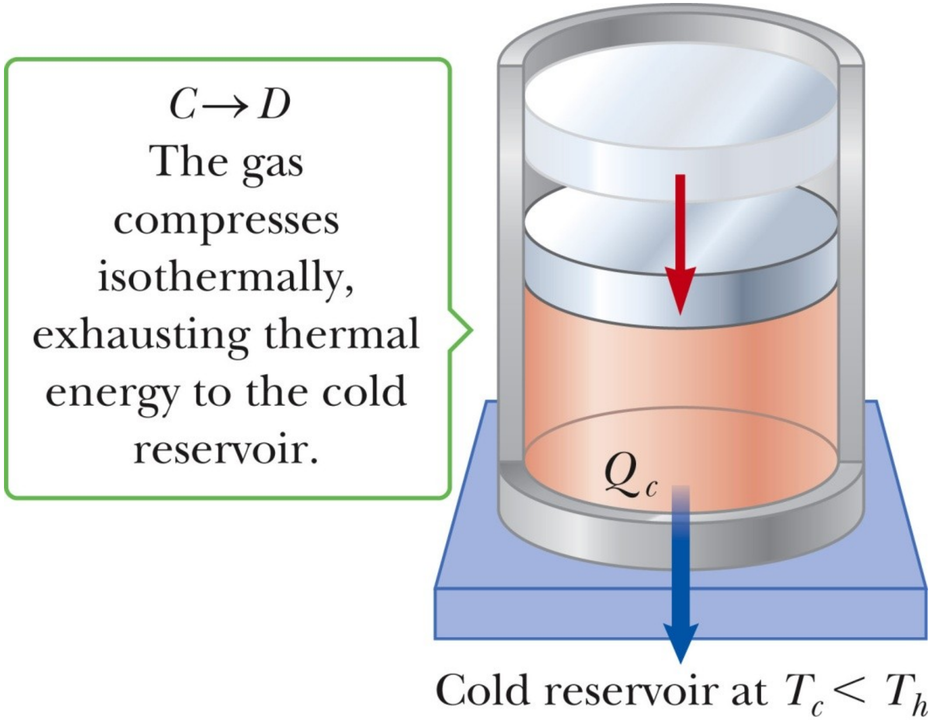
Carnot Cycle, B to C

- B to C is an adiabatic expansion
- The base of the cylinder is replaced by a thermally nonconducting wall
- No heat enters or leaves the system
- The temperature falls from T_h to T_c
- The gas does work W_{BC}



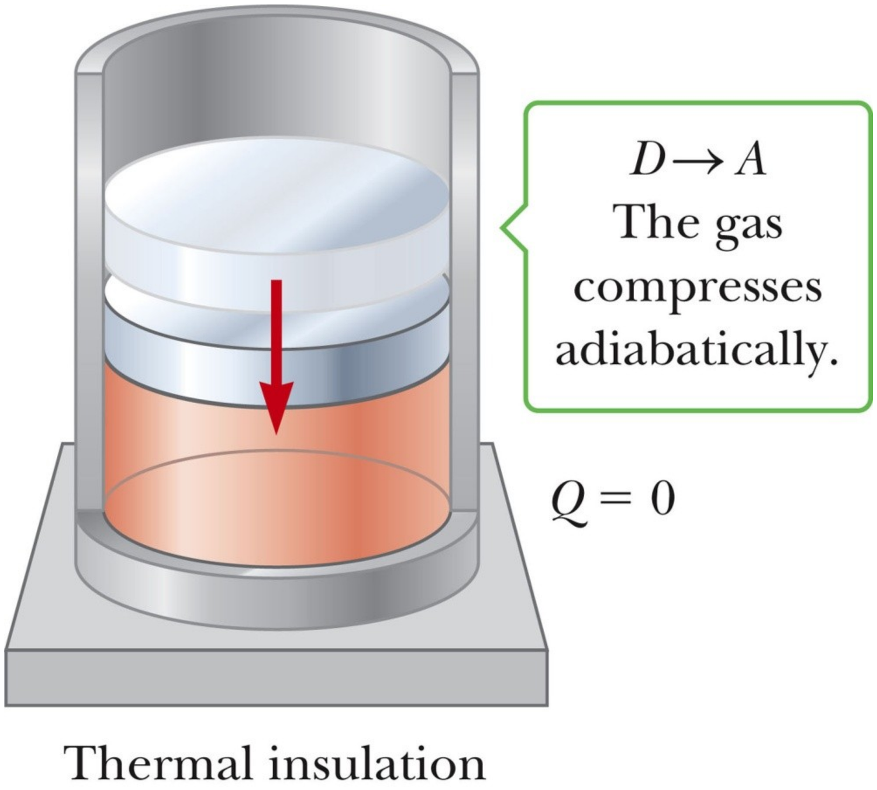
Carnot Cycle, C to D

- The gas is placed in contact with the cold temperature reservoir at temperature T_c
- C to D is an isothermal compression
- The gas expels energy Q_C
- Work W_{CD} is done on the gas



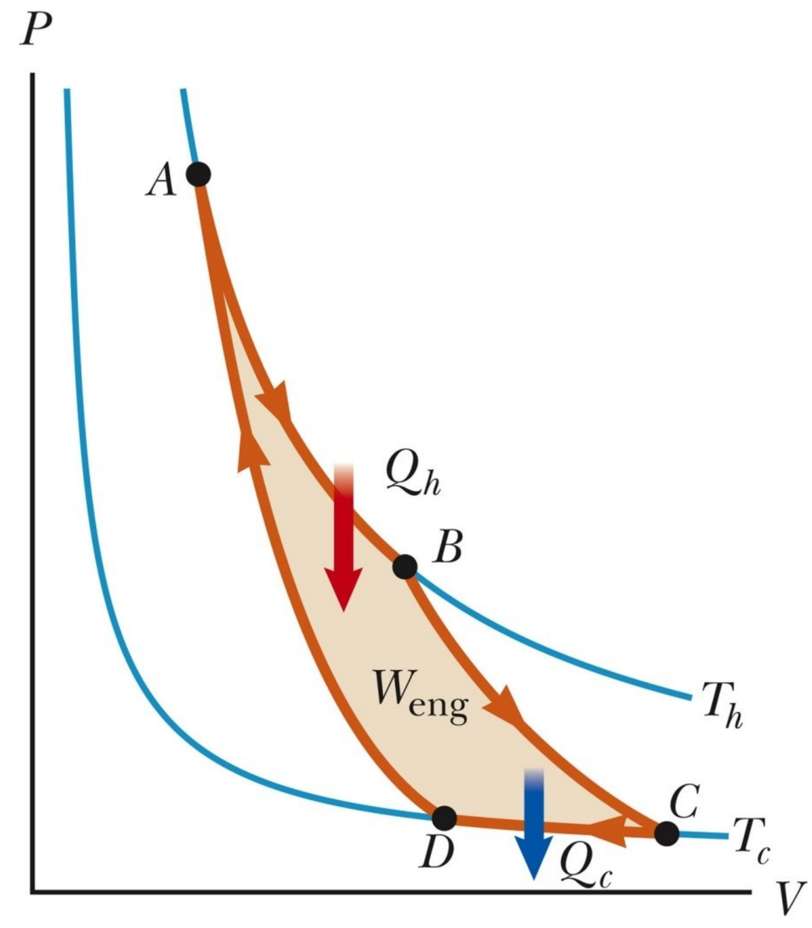
Carnot Cycle, D to A

- D to A is an adiabatic compression
- The gas is again placed against a thermally nonconducting wall
  - So no heat is exchanged with the surroundings
- The temperature of the gas increases from T_C to T_h
- The work done on the gas is W_{DA}



Carnot Cycle, PV Diagram

- The work done by the engine is shown by the area enclosed by the curve
- The net work, W_{eng}, is equal to $|Q_h|$ - $|Q_c|$



Efficiency of a Carnot Engine

- Carnot showed that the efficiency of the engine depends on the temperatures of the reservoirs

$$e_c = 1 - \frac{T_c}{T_h}$$

- Temperatures must be in Kelvins
- All Carnot engines operating reversibly between the same two temperatures will have the same efficiency

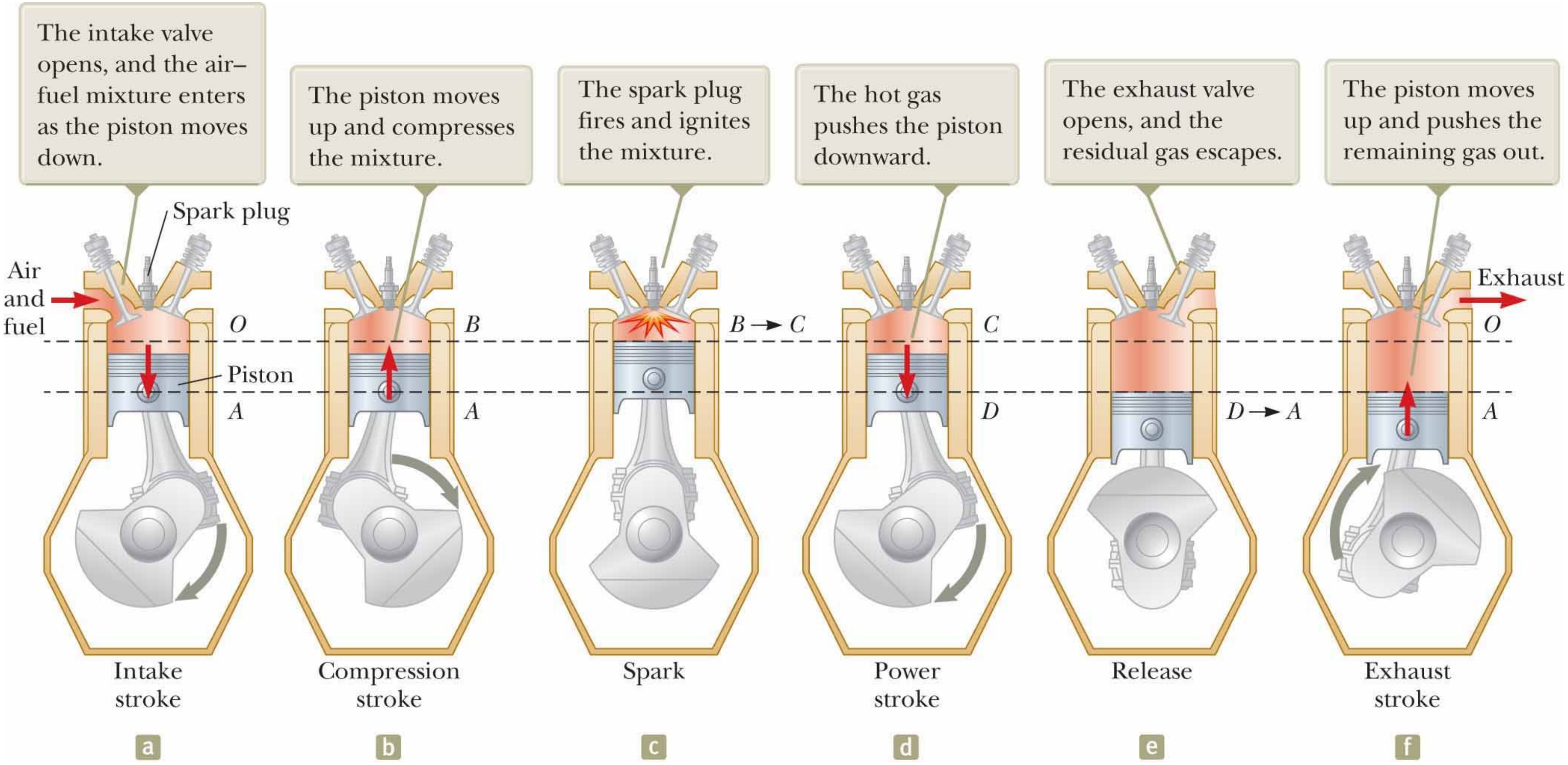
Notes About Carnot Efficiency

- Efficiency is 0 if $T_h = T_c$
- Efficiency is 100% only if $T_c = 0$ K
  - Such reservoirs are not available
    - Due to the Third Law of Thermodynamics
- The efficiency increases as T_c is lowered and as T_h is raised
- In most practical cases, T_c is near room temperature, 300 K
  - So generally T_h is raised to increase efficiency

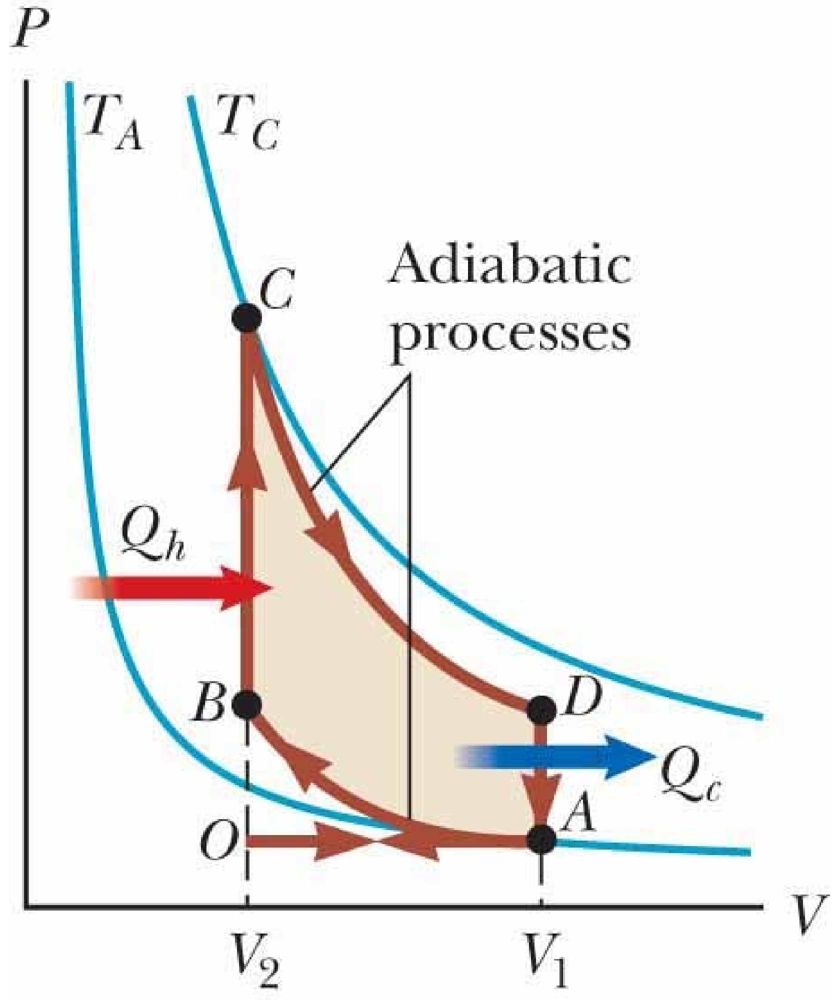
Real Engines Compared to Carnot Engines

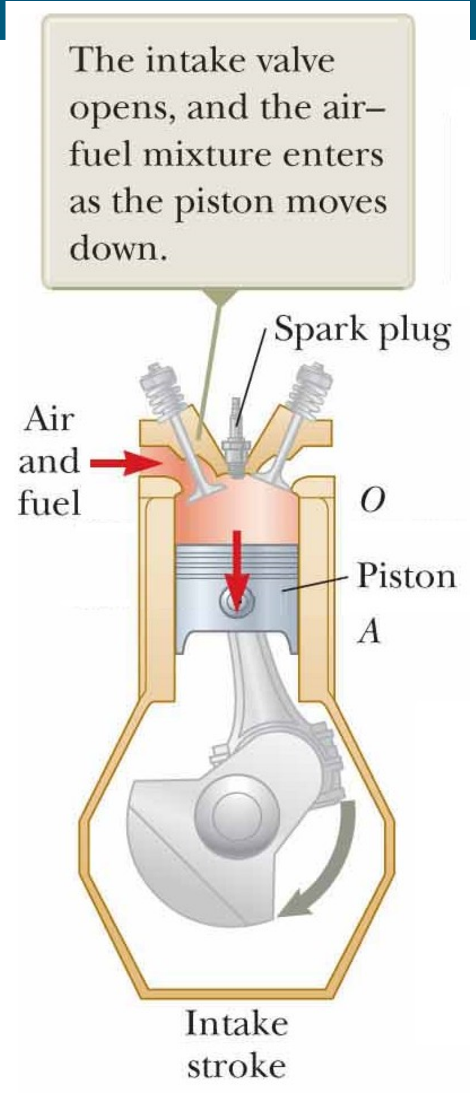
- All real engines are less efficient than the Carnot engine
  - Real engines are irreversible because of friction
  - Real engines are irreversible because they complete cycles in short amounts of time

Gasoline and Diesel Engines



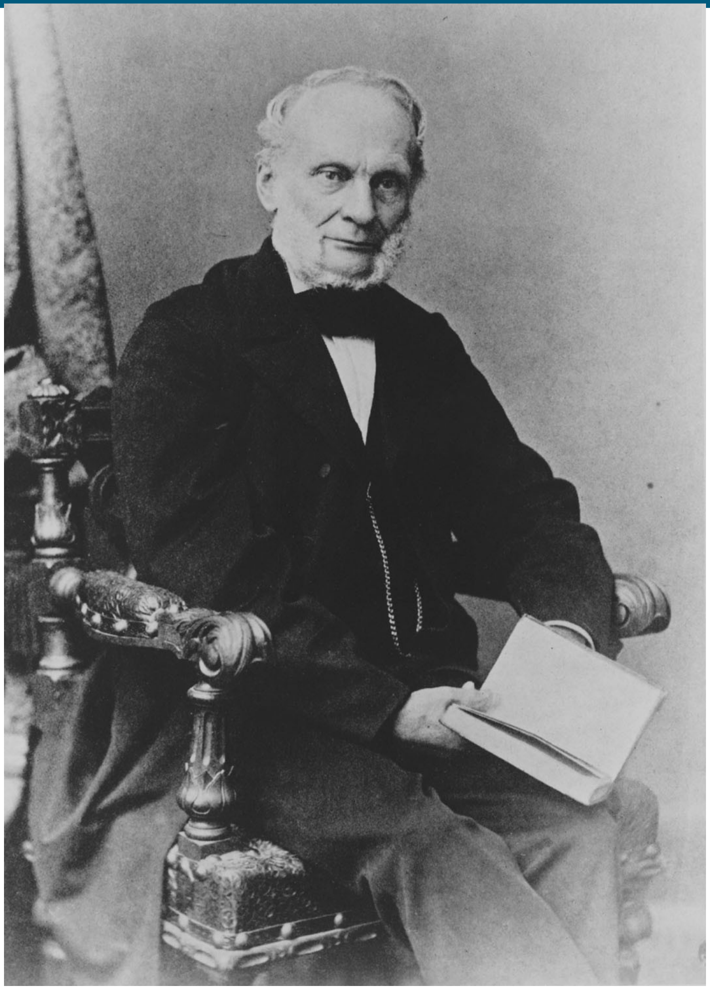
The Otto Cycle: Intake Stroke





Rudolf Clausius

- 1822 – 1888
- German physicist
- Ideas of entropy



Entropy

- A state variable related to the Second Law of Thermodynamics, the entropy
- Let Q_r be the energy absorbed or expelled during a reversible, constant temperature process between two equilibrium states
  - Then the change in entropy during any constant temperature process connecting the two equilibrium states can be defined as the ratio of the energy to the temperature
  - https://www.youtube.com/watch?v=ykUmibZHEZk

Entropy, cont.

- Mathematically, $\Delta S = \frac{Q_r}{T}$
  - SI unit: J/K
- This applies only to the reversible path, even if the system actually follows an irreversible path
  - To calculate the entropy for an irreversible process, model it as a reversible process
- The entropy change calculated for the reversible path is taken to be ΔS for the actual path
- The change in entropy depends only on the endpoints, not on the path connecting them

More About Entropy

- Note, the equation defines the *change in entropy*
- The entropy of the Universe increases in all natural processes
  - This is another way of expressing the Second Law of Thermodynamics
- There are processes in which the entropy of a system decreases
  - If the entropy of one system, A, decreases it will be accompanied by the increase of entropy of another system, B
  - The change in entropy in system B will be greater than that of system A

Perpetual Motion Machines

- A perpetual motion machine would operate continuously without input of energy and without any net increase in entropy
- Perpetual motion machines of the first type would violate the First Law, giving out more energy than was put into the machine
- Perpetual motion machines of the second type would violate the Second Law, possibly by no exhaust
- Perpetual motion machines will never be invented

Entropy and Disorder

- Entropy can be described in terms of disorder
- A disorderly arrangement is much more probable than an orderly one if the laws of nature are allowed to act without interference
  - This comes from a statistical mechanics development

Entropy and Disorder, cont.

- Isolated systems tend toward greater disorder, and entropy is a measure of that disorder
  - $S = k_B \ln W$
    - k_B is Boltzmann's constant
    - W is a number proportional to the probability that the system has a particular configuration
- This gives the Second Law as a statement of what is most probable rather than what must be
- The Second Law also defines the direction of time of all events as the direction in which the entropy of the universe increases

Grades of Energy

- The tendency of nature to move toward a state of disorder affects a system's ability to do work
- Various forms of energy can be converted into internal energy, but the reverse transformation is never complete
- If two kinds of energy, A and B, can be completely interconverted, they are of the *same grade*

Grades of Energy, cont.

- If form A can be completely converted to form B, but the reverse is never complete, A is a *higher grade* of energy than B
- When a high-grade energy is converted to internal energy, it can never be fully recovered as high-grade energy
- *Degradation of energy* is the conversion of high-grade energy to internal energy
- In all real processes, the energy available for doing work decreases

Heat Death of the Universe

- The entropy of the Universe always increases
- The entropy of the Universe should ultimately reach a maximum
  - At this time, the Universe will be at a state of uniform temperature and density
  - This state of perfect disorder implies no energy will be available for doing work
- This state is called the *heat death* of the Universe

The First Law and Human Metabolism

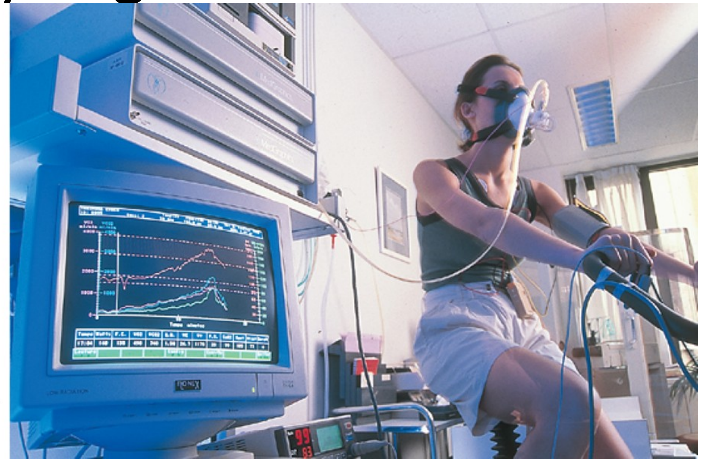
- The First Law can be applied to living organisms
- The internal energy stored in humans goes into other forms needed by the organs and into work and heat
- The *metabolic rate* (ΔU / Δt) is the rate at which chemical potential energy in food and oxygen are transformed into internal energy to just balance the body losses of internal energy by work and heat

Measuring Metabolic Rate

- The metabolic rate is directly proportional to the rate of oxygen consumption by volume

$$\frac{\Delta U}{\Delta t} = 4.8 \frac{\Delta V_{O_2}}{\Delta t}$$

- About 80 W is the *basal metabolic rate*, just to maintain and run different body organs



Laurent/B./American Hospital of Paris/Science Source

Various Metabolic Rates

Table 12.4 Oxygen Consumption and Metabolic Rates for Various Activities for a 65-kg Male[a]

Activity	O_2 Use Rate (mL/min · kg)	Metabolic Rate (kcal/h)	Metabolic Rate (W)
Sleeping	3.5	70	80
Light activity (dressing, walking slowly, desk work)	10	200	230
Moderate activity (walking briskly)	20	400	465
Heavy activity (basketball, swimming a fast breaststroke)	30	600	700
Extreme activity (bicycle racing)	70	1 400	1 600

[a] *Source: A Companion to Medical Studies*, 2/e, R. Passmore, Philadelphia, F. A. Davis, 1968.

Aerobic Fitness

- One way to measure a person's physical fitness is their maximum capacity to use or consume oxygen

$$\frac{\Delta U}{\Delta t} = \frac{Q}{\Delta t} + \frac{W}{\Delta t}$$

e = body's efficiency

$$= \frac{|W/\Delta t|}{|\Delta U/\Delta t|}$$

Table 12.5 Physical Fitness and Maximum Oxygen Consumption Rate[a]

Fitness Level	Maximum Oxygen Consumption Rate (mL/min · kg)
Very poor	28
Poor	34
Fair	42
Good	52
Excellent	70

[a] *Source: Aerobics*, K. H. Cooper, Bantam Books, New York, 1968.

Efficiency of the Human Body

Table 12.6 Metabolic Rate, Power Output, and Efficiency for Different Activities[a]

Activity	Metabolic Rate $\frac{\Delta U}{\Delta t}$ (watts)	Power Output $\frac{W}{\Delta t}$ (watts)	Efficiency e
Cycling	505	96	0.19
Pushing loaded coal cars in a mine	525	90	0.17
Shoveling	570	17.5	0.03

[a] *Source:* "Inter- and Intra-Individual Differences in Energy Expenditure and Mechanical Efficiency," C. H. Wyndham et al., *Ergonomics* **9,** 17 (1966).

- Efficiency is the ratio of the mechanical power supplied to the metabolic rate or total power input